

Radians



1

a 0.56

b 0.83

c 0.83

d 1.48

e -4.38

f -0.50

g 0.87

h -0.81

i -0.59

Exact values

2

a $h = 1$ unit

b

i $\frac{\sqrt{3}}{2}$

ii $\frac{1}{2}$

iii $\sqrt{3}$

iv $\frac{1}{2}$

3

a $h = \sqrt{2}$ units

b

i $\frac{\sqrt{2}}{2}$

ii $\frac{\sqrt{2}}{2}$

iii 1

4

a 1

b 0

c 0

d 1

e 0

5

a $\frac{\sqrt{3} + 1}{2}$

b $\frac{\sqrt{2}}{4}$

c 1

d $\frac{\sqrt{6}}{4} + 1$

e 0

f 2

Exact values from reference angles

6 $\alpha = \frac{\pi}{11}$

7 $\alpha = \frac{\pi}{7}$

8 $\alpha = \frac{\pi}{5}$

9

a $\frac{1}{2}$

b -1



c $-\frac{1}{2}$

d $-\frac{\sqrt{3}}{2}$

e $-\frac{\sqrt{3}}{2}$

f $\frac{1}{2}$

g 1

h 0

i 1

j 0

k $-\frac{\sqrt{2}}{2}$

l $-\frac{\sqrt{2}}{2}$

m $-\frac{\sqrt{2}}{2}$

n 1

o $\frac{\sqrt{3}}{3}$

p $-\frac{\sqrt{3}}{3}$

10

a $-\frac{1}{2}$

b $-\frac{\sqrt{3}}{2}$

c $-\frac{1}{2}$

d $\frac{\sqrt{3}}{3}$

11

a $-\frac{\sqrt{3}}{4}$

b -2

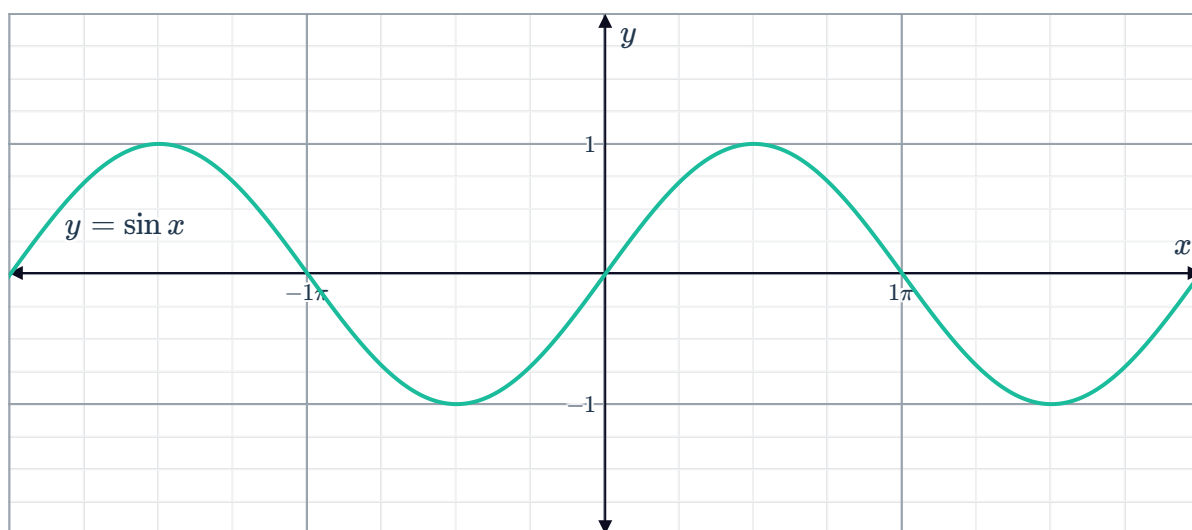
Trigonometric functions

12

a

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

b

c 0

d Negative

e Negative

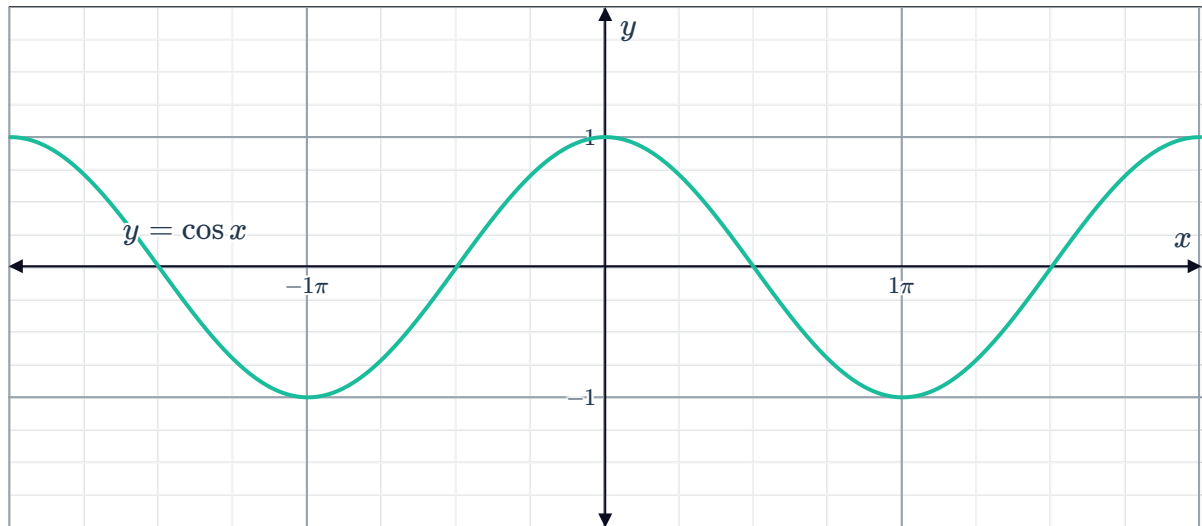


13

a

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c -1

d Positive

e Positive

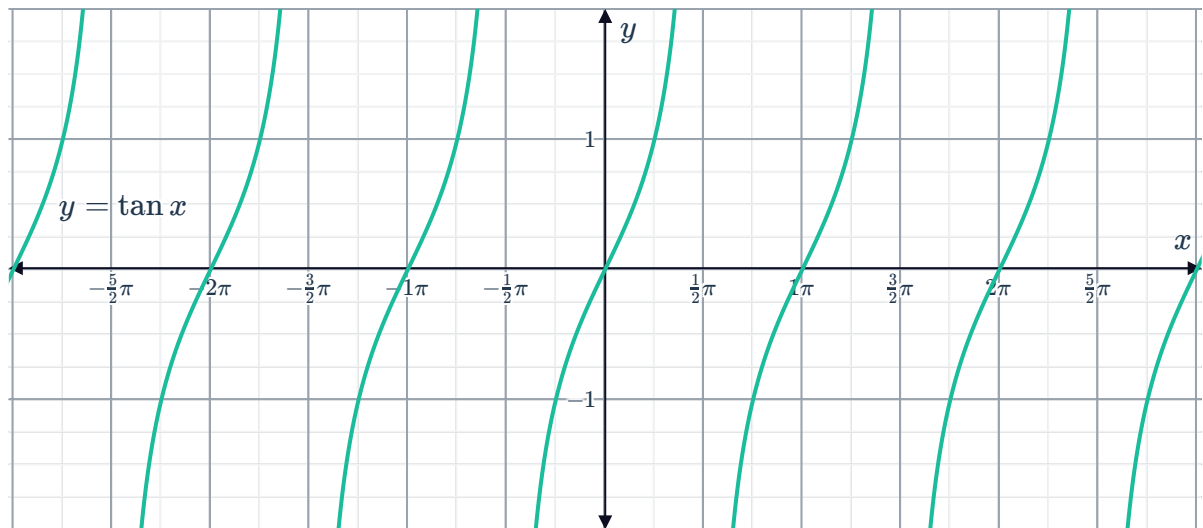
f Quadrant 4



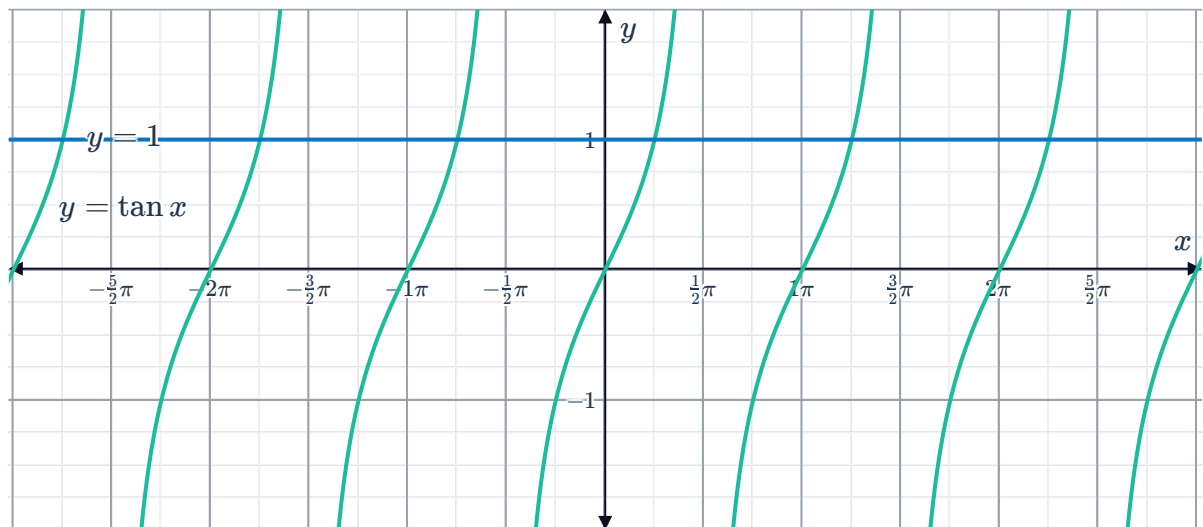
a

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\tan x$	0	1	undefined	-1	0	1	undefined	-1	0

b



c



d $x = -\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$

e 0

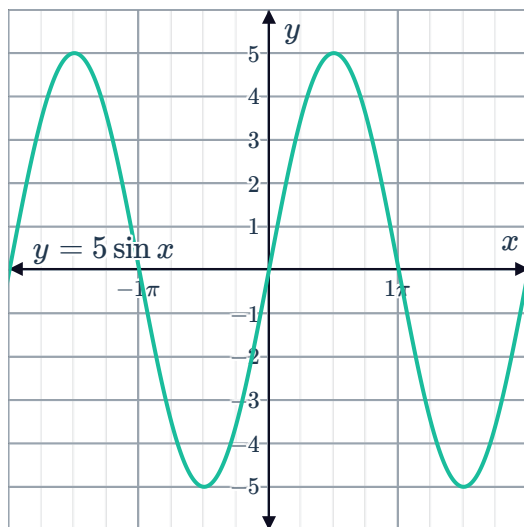
f Negative

g Negative

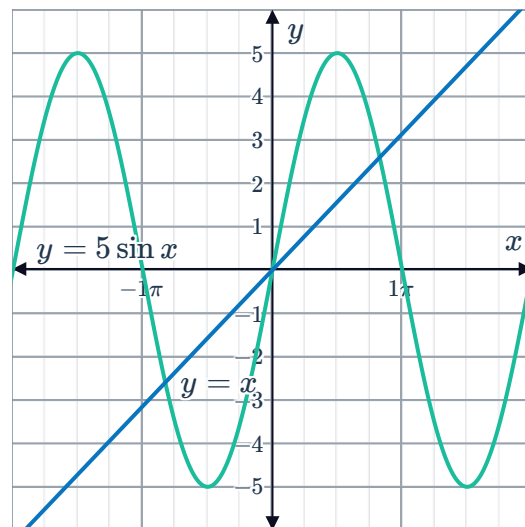
h Quadrant 4

15

a



b



c 3 solutions

16 Point C